PHARMACOVIGILANCE

Important Topics • Step-by-Step Study Notes

Beginner guide for B.Pharm students and Pharmacovigilance fresher preparation

Is PDF ka use kaise karein?
Topics ko diye gaye order mein padhein. Har step ke baad short revision karein aur end mein diya gaya practice case solve karein. Language simple Hinglish rakhi gayi hai; important technical terms English mein hain.

Note: Yeh video ke subject (ADR aur Pharmacovigilance) par based study guide hai. Video ka complete transcript verify nahi ho saka, isliye ise exact line-by-line video summary na samjhein.

STEP 1 — Pharmacovigilance ka meaning

Pharmacovigilance (PV) medicines ke adverse effects aur medicine-related problems ko detect, assess, understand aur prevent karne ki science aur activities hain. Simple words mein: medicine patient ke liye safe hai ya nahi, uski monitoring karna.

Main objectives: (1) patient safety improve karna, (2) suspected side effects identify karna, (3) medicine ke risks ko understand karna, (4) harm prevent karna, aur (5) medicine ka safe use promote karna.

STEP 2 — Adverse Event (AE) aur Adverse Drug Reaction (ADR)

Adverse Event (AE): medicine use karne ke baad koi unwanted medical occurrence ho. Iska medicine se cause-and-effect relation zaroori nahi hota.

Adverse Drug Reaction (ADR): medicine ke normal/approved use ke dauran hone wala harmful ya unintended response, jisme medicine ke saath kam-se-kam reasonable possibility of causal relationship hoti hai.

AE	ADR
Medicine ke baad unwanted event; cause unknown ho sakta hai.	Medicine ke saath suspected/possible causal relationship hota hai.
Example: medicine lete waqt patient ko fever ho gaya; cause abhi clear nahi.	Example: medicine ke baad rash aaya aur assessment mein medicine suspected hai.

STEP 3 — Types of ADR (A to F)

Type	Name	Meaning / clue
A	Augmented	Dose-related; known pharmacological action se predictable ho sakta hai.
B	Bizarre	Unpredictable; usual dose par bhi ho sakta hai.
C	Chronic	Long-term use se associated.
D	Delayed	Reaction kuch time baad appear hota hai.
E	End of use	Medicine stop/withdraw karne ke baad hota hai.
F	Failure	Therapy expected effect nahi deti; treatment failure.

STEP 4 — Seriousness vs Severity

Severity batata hai reaction kitna intense hai: mild, moderate ya severe. Seriousness regulatory criteria par based hoti hai. Serious outcome mein death, life-threatening event, hospitalization/prolongation, significant disability, congenital anomaly, ya medically important event aa sakta hai.

Yaad rakhein: Severe reaction hamesha regulatory sense mein serious nahi hota; seriousness aur severity alag concepts hain.

STEP 5 — Expected vs Unexpected ADR

Expected: reaction applicable product information/reference safety information mein already described ho.
Unexpected: reaction ki nature, severity ya specificity available reference safety information se consistent na ho.

STEP 6 — ICSR (Individual Case Safety Report)

ICSR ek individual report hoti hai jo ek patient mein suspected adverse reaction aur suspected medicine ki details record karti hai. Pharmacovigilance case processing ka yeh core document hai.

Valid ICSR ke 4 minimum elements:

1. Identifiable patient
2. Identifiable reporter
3. Suspected medicinal product
4. Suspected adverse reaction/event

STEP 7 — Case processing workflow

No.	Stage	What happens?
1	Intake / receipt	Report receive hoti hai (patient, doctor, pharmacist ya other source se).
2	Validity check	4 minimum ICSR elements present hain ya nahi, check karein.
3	Duplicate check	Same case pehle report hua hai ya nahi, compare karein.
4	Data entry	Patient, reporter, medicine, event, dates, history aur relevant details enter karein.
5	Coding	Reaction ko standard medical terminology (e.g. MedDRA) se code kiya ja sakta hai.
6	Medical review	Clinical details, seriousness, expectedness aur causality review hoti hai.
7	Quality check	Accuracy, completeness, consistency aur documentation verify karein.
8	Submission / follow-up	Applicable timelines aur rules ke mutabiq report submit/follow-up hota hai.

STEP 8 — Causality Assessment

Causality assessment mein evaluate kiya jata hai ki suspected medicine aur reaction ke beech relation kitna likely hai. Reviewer timing, known drug information, alternative causes, medicine stop karne ke baad kya hua (dechallenge), aur zaroorat par restart ke baad kya hua (rechallenge) jaise factors consider kar sakta hai.

Important: causality hamesha certainty prove nahi karti. Case ke available evidence ke basis par assessment hota hai.

STEP 9 — MedDRA basics

MedDRA (Medical Dictionary for Regulatory Activities) ek standardized medical terminology hai. Iska use adverse events/reactions ko consistent way mein code aur classify karne ke liye hota hai. Fresher level par iska purpose aur basic hierarchy samajhna start karein.

STEP 10 — WHO-UMC aur PvPI

WHO-UMC: suspected medicine-reaction relationship ki causality assessment ke liye commonly used system/criteria.
PvPI: Pharmacovigilance Programme of India, jo India mein medicines ki safety monitoring aur ADR reporting ko promote karta hai. Current reporting channels aur guidance ke liye official PvPI/IPC information check karein.

STEP 11 — Signal detection and risk management

Signal: safety information ka aisa observation jo kisi medicine aur event ke beech nayi ya changed possible association ki taraf attention dilaye; further evaluation zaroori hoti hai.
Risk management: medicine ke known/potential risks ko understand aur minimize karne ke measures.

STEP 12 — Practical practice case

Scenario: Patient ne prescribed medicine use ki. Kuch waqt baad skin rash report hua. Abhi cause confirm nahi hai.

Practice tasks:
1. Kya yeh AE report hai, ADR hai, ya suspected ADR? Reason likhein.
2. ICSR ke 4 minimum elements list karein.
3. Reporter se kaunsi additional information leni chahiye?
4. Event ki start date, medicine start date, dose aur medical history kyun important hain?
5. Seriousness aur severity ko alag-alag assess karne ki zaroorat kyun hai?

Model learning point: Sirf rash aane se medicine ko confirmed cause nahi keh sakte. Isse suspected reaction ke roop mein document karke complete information aur medical assessment collect ki jati hai.

Fresher ke liye study order / revision checklist

☐ Pharmacovigilance definition and objectives

☐ AE vs ADR

☐ ADR classification (A–F)

☐ Seriousness vs severity

☐ Expected vs unexpected reaction

☐ ICSR and its 4 minimum elements

☐ Case processing workflow

☐ Causality assessment basics

☐ MedDRA basics

☐ WHO-UMC and PvPI basics

☐ Signal detection and risk management

Quick glossary

Term	Simple meaning
ADR	Adverse Drug Reaction — suspected harmful reaction related to a medicine.
AE	Adverse Event — unwanted medical occurrence; cause may be unrelated/unknown.
ICSR	Individual Case Safety Report.
MedDRA	Standard medical terminology used for coding safety information.
Dechallenge	Observation after stopping/reducing suspected medicine.
Rechallenge	Observation after restarting suspected medicine, if it occurs.
Signal	Information suggesting a possible new or changed safety concern.

Study tip: Pehle Steps 1–7 strong karein. Entry-level PV interviews mein definitions, AE vs ADR, seriousness vs severity, ICSR ke 4 elements aur case-processing sequence ko clearly explain karna useful hota hai. Yeh preparation guide hai, official regulatory SOP ka substitute nahi.

End of notes